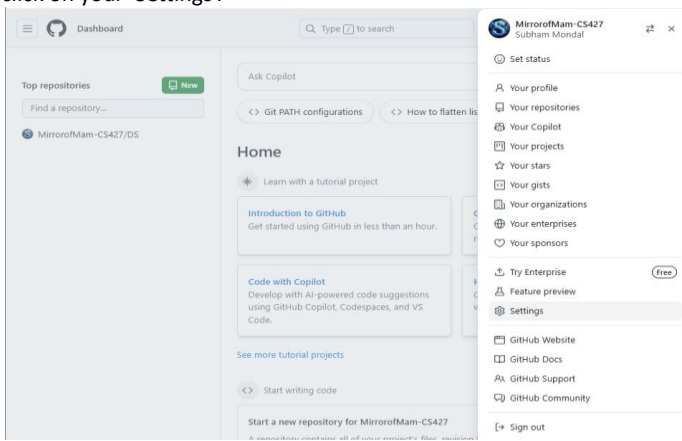
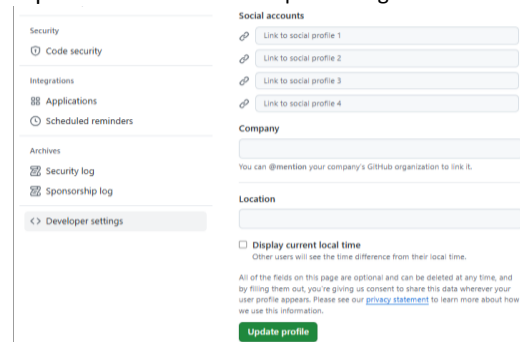


Assignment 9: Deploy a project from GitHub to EC2.

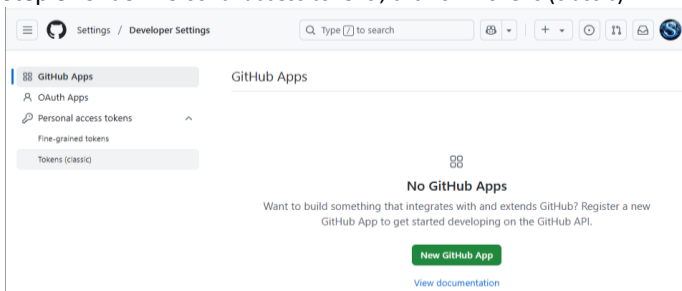
Step 1: After login into GitHub account, go to your profile section and click on your 'Settings'.



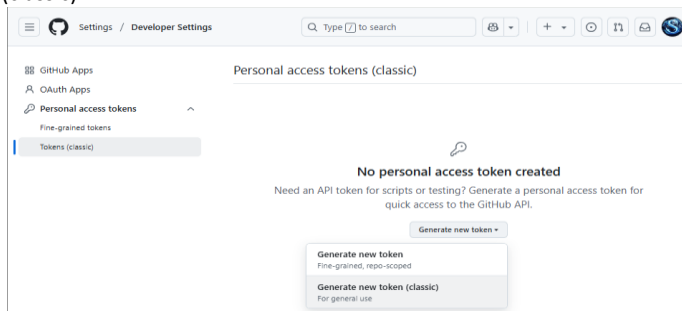
Step 2: Now click on 'Developer settings'.



Step 3: Under 'Personal access tokens', click on 'Tokens (classic)'.

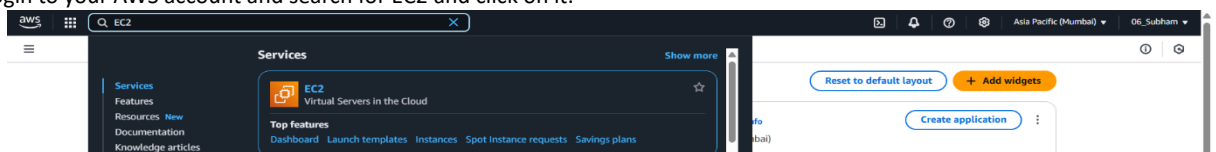


Step 4: Under 'Generate new token', click on 'Generate new token (classic)'.

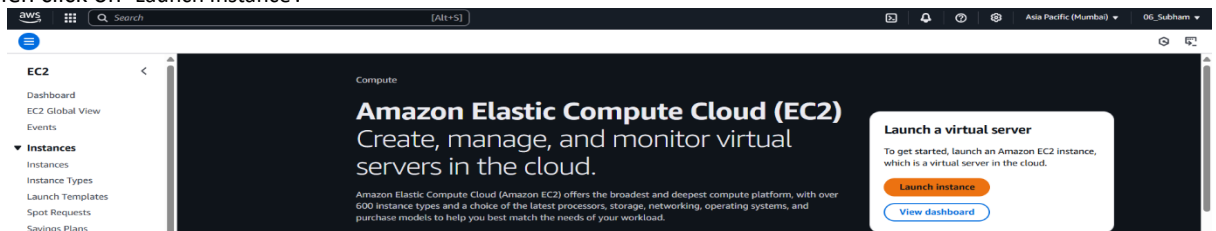


Note: After that upload and commit a .js file along with .json file (index.js and Package.json).

Step 7: Login to your AWS account and search for EC2 and click on it.



Step 8: Then click on 'Launch Instance'.



Step 5: Give a suitable Note, then tick on the check boxes then click on 'Generate token'.



Step 9: Scroll down to 'Key pair (login)' and click on 'Create new key pair'. A dialog box appears, fill the box as follows.

The screenshot shows the AWS 'Launch an instance' page. The 'Key pair (login)' section is expanded, and the 'Create new key pair' button is clicked. A 'Create key pair' dialog box is open, showing the following details:

- Key pair name:** keyG7
- Key pair type:** RSA (selected), ED25519
- Private key file format:** pem (selected), pk8
- Instructions:** When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance.

Step 10: Give a Name of the Instance and select Ubuntu as the OS and allow the necessary settings as follows. Then click on 'Launch Instance'.

The screenshot shows the AWS 'Launch an instance' page with the following settings:

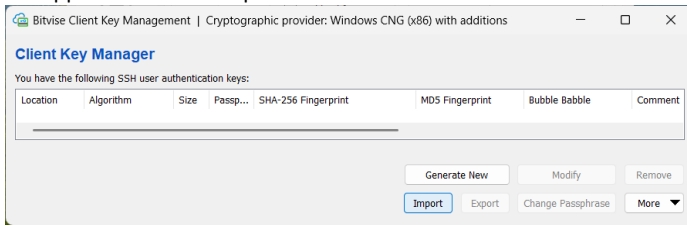
- Name and tags:** Name: G7_server
- Application and OS Images (Amazon Machine Image):** Ubuntu Server 24.04 LTS (HVM), SSD Volume Type (ami-0e35ddab05955cf57)
- Instance type:** t2.micro
- Key pair (login):** keyG7
- Network settings:** vpc-02493897f3c47f5e0, Subnet: No preference, Auto-assign public IP: Enable, Firewall (security groups): Create security group, Allow SSH traffic from: Anywhere, Allow HTTPS traffic from the internet, Allow HTTP traffic from the internet.
- Configure storage:** 1x 8 GiB gp3, Root volume, 3000 IOPS, Not encrypted.

The 'Launch Instance' button is visible at the bottom right.

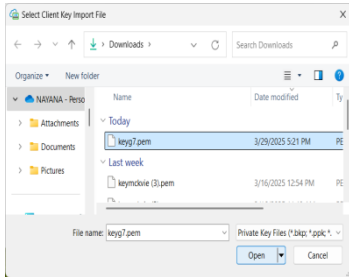
Note: Click on the check box of the instance in the dashboard section, the details of the instance is visible.

Note: Now to create a connection between the instance (server) and the client, we need to use a third-party application Bitvise SSH Client. After installing the app a window will appears (like step 15 or 17).

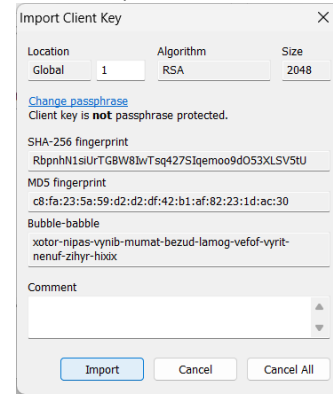
Step 11: From that click on 'Client key manager'. The following dialog box appears. Click on 'Import'.



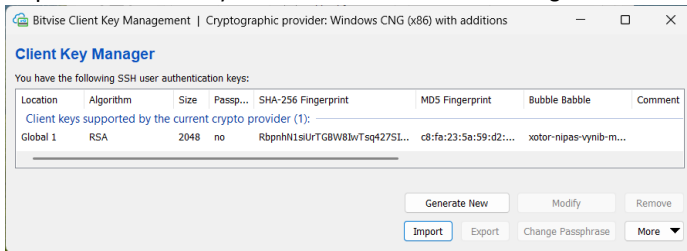
Step 12: Now select the file containing the client key and click on 'Open'.



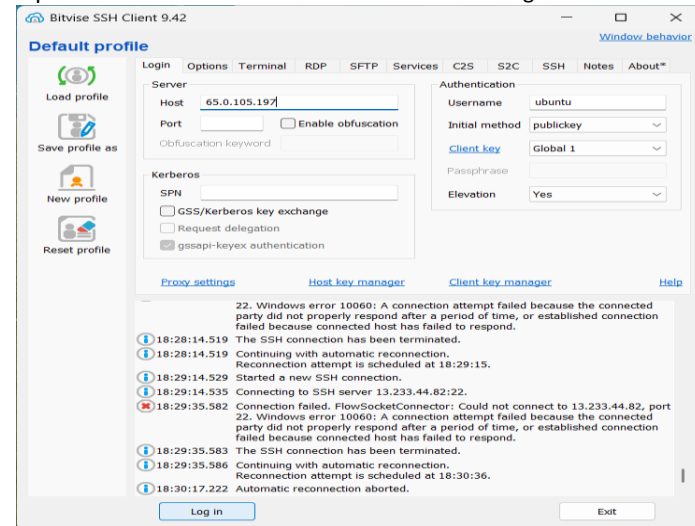
Step 13: The following appears, click on 'Import'.



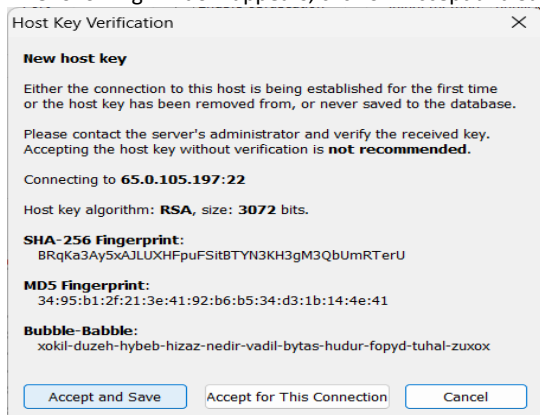
Step 14: The client key is now visible. Close the following window.



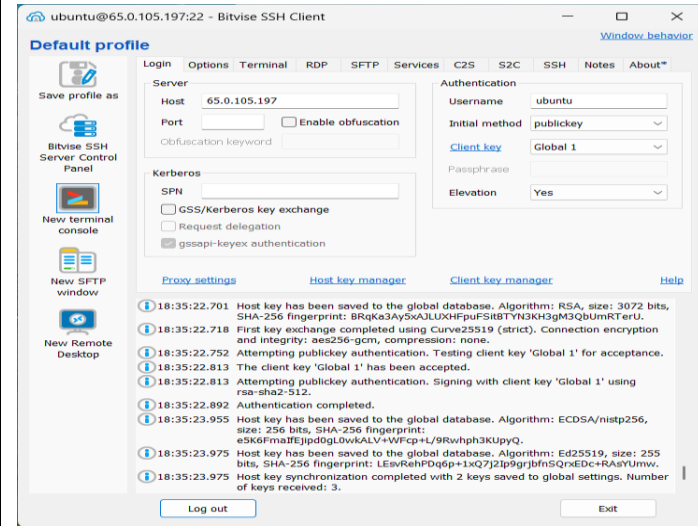
Step 15: Now fill the details as follows and click on login.



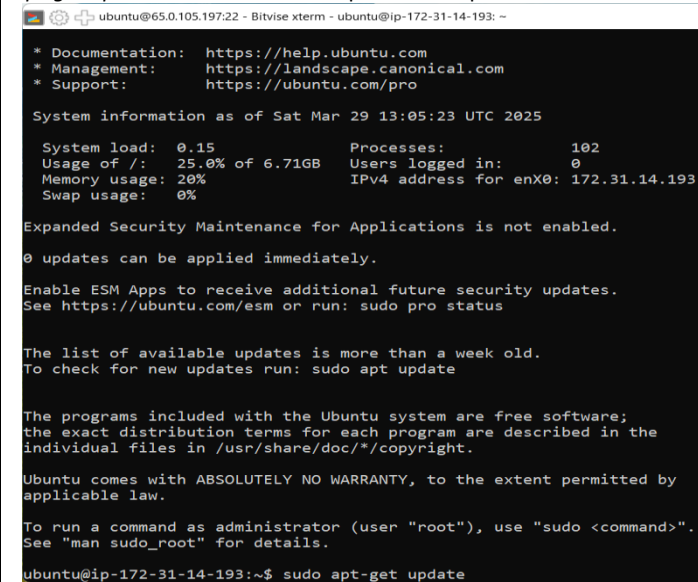
Step 16: The following window appears, click on 'Accept and Save'.



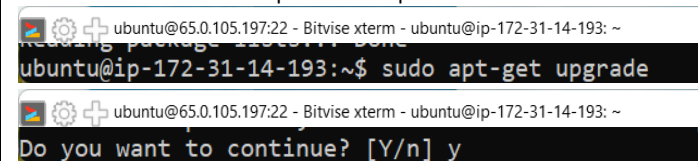
Step 17: The login process is successful. Now click on 'New terminal console' from the below window.



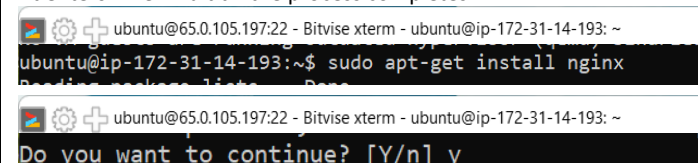
Step 18: Then a terminal will be open and run the command "sudo apt-get update". Then wait till the process completes.



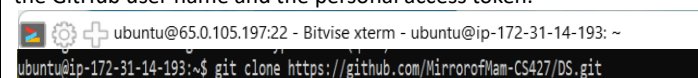
Step 19: Run the command "sudo apt-get upgrade". Then type yes the hit enter. Then wait till the process completes.



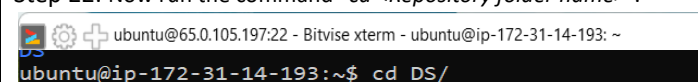
Step 20: Run the command "sudo apt-get install nginx". Type yes, the hit enters. Then wait till the process completes.



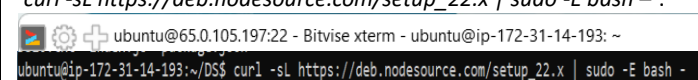
Step 21: Now run the command "git clone <Repository link>". In case of private GitHub repository, we need to login into the terminal using the GitHub user name and the personal access token.



Step 22: Now run the command "cd <Repository folder name>".



Step 23: Now run the command "curl -sL https://deb.nodesource.com/setup_22.x | sudo -E bash -".



Step 24: Now run the command “*sudo apt install nodejs*”.

```
ubuntu@ip-172-31-14-193:~/DS$ sudo apt install nodejs
```

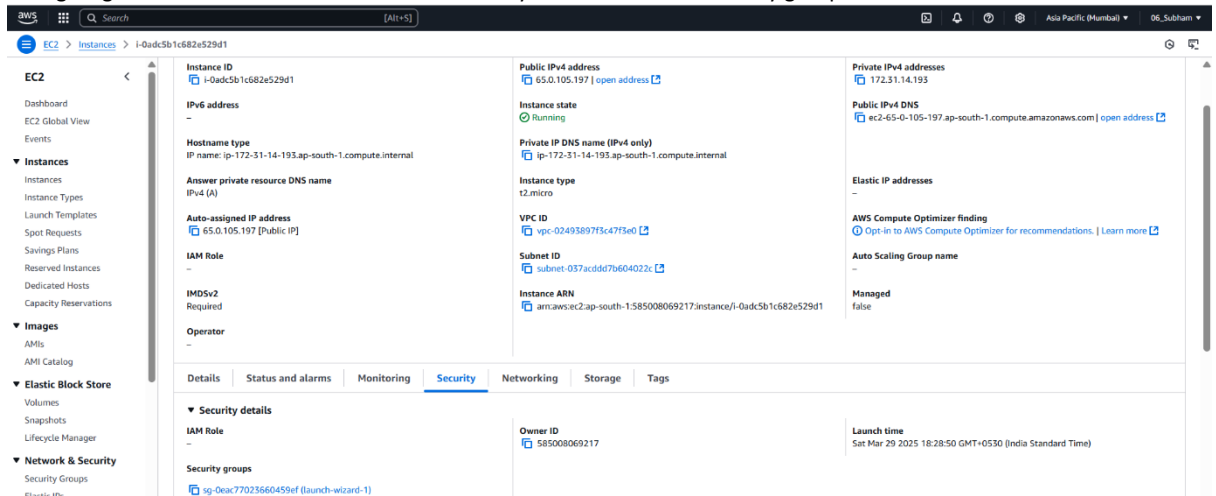
Step 25: Now run the command “*npm install*”.

```
ubuntu@ip-172-31-14-193:~/DS$ npm install
```

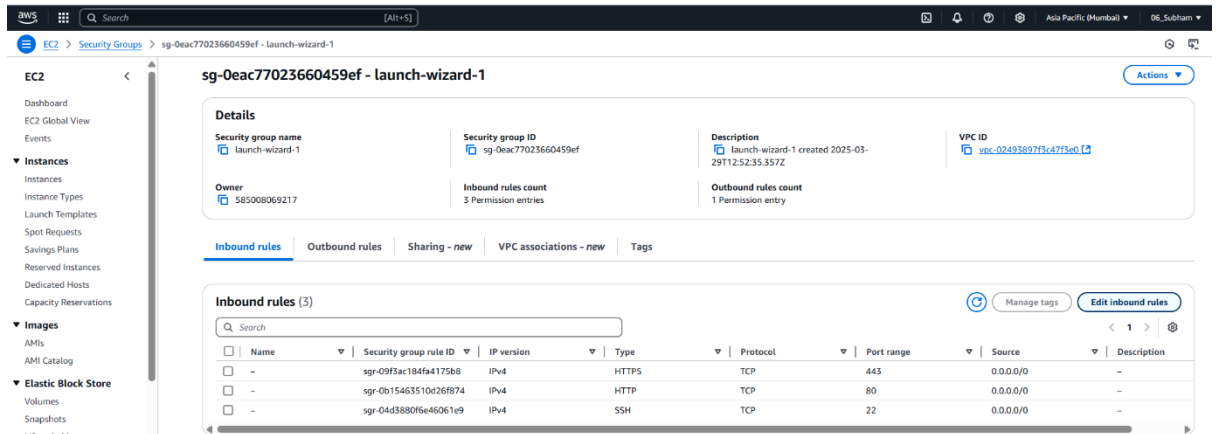
Step 26: Now give the command “*node <.js file>*” and press enter. The Server gets started.

```
ubuntu@ip-172-31-14-193:~/DS$ node index.js
Started server
```

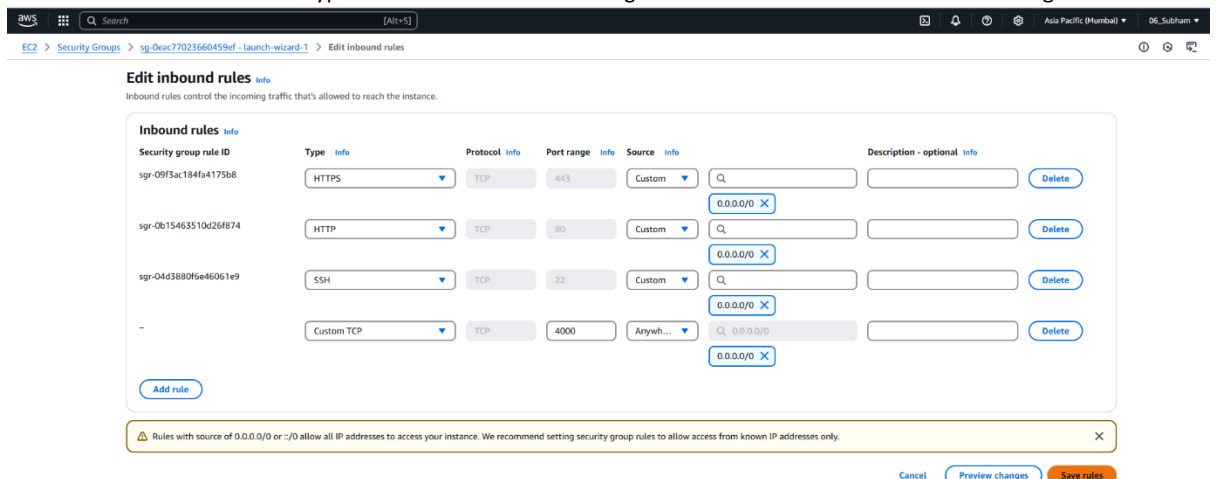
Step 27: Now again go to the EC2 instance and click on ‘Security’. Then under the Security group click on the blue coloured line.



Step 28: Click on ‘Edit inbound rules’.



Step 29: Click on ‘Add rule’. Give the rule type of “Custom TCP” & Port range at 4000 and rest of the fill as bellow image and click on ‘Save rules’.



Step 30: Open a browsing window and give the “<instance IP address>:4000” and hit enter. The web page (.js file) is seen.

